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TECHNICAL PSEUDOCODE

Pseudocode & Technical MCQ Master E-Book

Execution Tracing, Loops, Bitwise Operators, Algorithms & Output Predictions

Target Role: Infosys SE (2026 Batch)

Total Questions: 41 Solved MCQs

Format: CampusMonk Master E-Book Standard

Status: 100% Verified & Answer-Keyed

Q001 PSEUDO CODE

How many times does the loop body run, and what is printed?

```
Integer
i, count
Set
i = 1, count = 0
while
i < 20 do
i
= i * 2
count
= count + 1
end
while
Print
count
```

(A) 3
(B) 4
(C) 5
(D) 6

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Trace i and count: start i=1. Pass 1: i=2, count=1. Pass 2: i=4, count=2. Pass 3: i=8, count=3. Pass 4: i=16, count=4. Pass 5: i=32, count=5. Now i=32 is not less than 20, so the loop stops. Printed value is 5.

Q002 PSEUDO CODE

What is the output of the following Pseudo Code? Integer division is used.

```
Integer
a, b
Set
a = 17, b = 5
Print
(a / b) + (a % b)
```

A 3
B 4
C 5
D 7

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Integer division: 17 / 5 = 3. Remainder: 17 % 5 = 2. Printed value = 3 + 2 = 5.

Q003 PSEUDO CODE

What is the output of the following Pseudo Code?

```
Integer
x, i
Set
x = 0
for
i = 1 to 6 do
if
i % 2 == 0 then
continue
end
if
x
= x + i
end
for
Print
x
```

A 9
B 12
C 16
D 21

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

continue skips even values of i. The loop runs i = 1 to 6, so only the odd values 1, 3 and 5 are added. x = 1 + 3 + 5 = 9.

Q004 PSEUDO CODE

What value is printed by the following Pseudo Code?

```
Function Mystery(n)
if n <= 1 then return 1
end if
return n * Mystery(n - 1)
End Function
Print Mystery(5)
```

A 15

B 24

C 120

D 125

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Mystery(n) computes n!. Mystery(5) = 5 × 4 × 3 × 2 × 1 = 120.

Q005 PSEUDO CODE

What is printed after the following Pseudo Code runs?

```
Integer
a, b
Set
a = 12, b = 8
a
= a + b
b
= a - b
a
= a - b
Print
a, b
```

A 12 8

B 8 12

C 20 8

D 20 12

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

This swaps the two values without a third variable. a = 12+8 = 20, b = 20-8 = 12, a = 20-12 = 8. Output: 8 12.

Q006 PSEUDO CODE

What is printed by the following Pseudo Code?

```
Integer
i, j, count
Set
count = 0
for
i = 1 to 4 do
for
j = 1 to i do
count
= count + 1
end
for
end
for
Print
count
```

A 4
B 8
C 10
D 16

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

The inner loop runs i times for each i from 1 to 4. Total iterations = 1+2+3+4 = 10. So count prints 10.

Q007 PSEUDO CODE

What is the output of the following Pseudo Code?

```
Integer
a, b
Set
a = 9, b = 5
Print
a ^ b
```

A 4
B 12
C 13
D 14

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

9 in binary is 1001 and 5 is 0101. XOR sets a bit where the bits differ: 1100, which is 12.

Q008 PSEUDO CODE

What value is printed by the following
Pseudo Code?

```
Function
F(n)
if
n == 0 then return 0
if
n == 1 then return 1
return
F(n - 1) + F(n - 2)
End
Function
Print
F(6)
```

A 5
B 8
C 13
D 21

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

F is the Fibonacci sequence: F(0)=0, F(1)=1,
F(2)=1, F(3)=2, F(4)=3, F(5)=5, F(6)=8. So the printed value is 8.

Q009 PSEUDO CODE

A queue follows FIFO. What is printed by the following Pseudo Code?

```
Queue
Q
enqueue(Q,
2)
enqueue(Q,
4)
enqueue(Q,
6)
dequeue(Q)
enqueue(Q,
8)
Print
front(Q)
```

A 2

B 4

C 6

D 8

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

After three enqueues the queue is [2, 4, 6]
with 2 at the front. dequeue removes 2. enqueue 8 makes [4, 6, 8].
front returns 4.

Q010 PSEUDO CODE

A stack follows LIFO. What is printed by the following Pseudo Code?
Stack
S
push(S,
5)
push(S,
8)
push(S,
3)
pop(S)
push(S,
7)
Print
peek(S)

A 3
B 5
C 7
D 8

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Stack after the three pushes: [5, 8, 3] with
3 on top. pop removes 3. push 7 makes the stack [5, 8, 7]. peek
returns the top element, which is 7.

Q011 PSEUDO CODE

What is printed?
Integer
n, d
Set
n = 48, d = 0
while
n > 1 do
n
= n / 2
d
= d + 1
end
while
Print
d

A 3
B 4
C 5
D 6

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
48→24→12→6→3→1.
That is five divisions. d = 5. (Integer division: 3/2=1.)

Q012 PSEUDO CODE

What is printed?
Integer
A[5] = {2, 4, 6, 8, 10}
Integer
i, s
Set
s = 0
for
i = 0 to 4 do
if
A[i] % 4 == 0 then s = s + A[i]
end
for
Print
s

A 12
B 18
C 24
D 30

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
Values divisible by 4 are 4 and 8. s = 4 + 8 = 12.

Q013 PSEUDO CODE

What is printed?
Integer
x
Set
x = 10
Print
x & 3

A 1
B 2
C 3
D 8

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
10 is 1010 and 3 is 0011. AND is 0010, which is 2.

Q014 PSEUDO CODE

What is printed?
Stack
S
push(S,
1)
push(S,
4)
push(S,
9)
pop(S)
pop(S)
Print
peek(S)

A 1
B 4
C 9
D Empty

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
After three pushes the top is 9. Two pops remove 9 then 4. peek returns 1.

Q015 PSEUDO CODE

What does G(4) return?
Function
G(n)
if
n <= 1 then return 1
return
n * G(n - 1)
End
Function
Print
G(4)

A 10
B 16
C 24
D 64

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
G is factorial. G(4)=4×3×2×1=24.

Q016 PSEUDO CODE

[Bitwise]
What is printed?
Integer
a
Set
a = 6
Print
a | 3

A 7
B 5
C 2
D 18

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
6
is 0110, 3 is 0011, OR is 0111 = 7.

Q017 PSEUDO CODE

[Queue]
What is printed?
Queue
Q
enqueue(Q,
11)
enqueue(Q,
22)
enqueue(Q,
33)
dequeue(Q)
dequeue(Q)
Print
front(Q)

A 11
B 22
C 33
D Empty

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Queue
starts [11,22,33]. Two dequeues remove 11 then 22. Front is 33.

Q018 PSEUDO CODE

[Loops]
What is printed?
Integer
i, s
Set
s = 0
for
i = 1 to 10 do
if
i % 2 == 1 then s = s + i
end
for
Print
s

A 25
B 30
C 45
D 55

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
The
loop adds the odd values 1+3+5+7+9 = 25.

Q019 PSEUDO CODE

[Recursion]
What does S(5) return?
Function
S(n)
if
n == 0 then return 0
return
n + S(n - 1)
End
Function
Print
S(5)

A 10
B 15
C 16
D 25

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
S(n)
is the sum 5+4+3+2+1+0 = 15.

Q020 PSEUDO CODE

[Arrays]
What is printed?
Integer
A[4] = {3, 5, 7, 9}
Integer
i, p
Set
p = 1
for
i = 0 to 3 do
p
= p * A[i]
end
for
Print
p

A 24
B 315
C 945
D 105

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Product
3×5×7×9 = 945.

Q021 PSEUDO CODE

[Loops]
What is printed?
Integer
i, p
Set
p = 1
for
i = 1 to 4 do
p
= p * i
end
for
Print
p

A 10
B 16
C 24
D 120

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
p
multiplies 1×2×3×4 = 24.

Q022 PSEUDO CODE

[Data
Structures] Queue Q receives 5, then 15, then 25. One dequeue
runs. What is front(Q)?
Queue
Q
enqueue(Q,
5)
enqueue(Q,
15)
enqueue(Q,
25)
dequeue(Q)
Print
front(Q)

A 5
B 15
C 25
D Empty

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
FIFO:
5 leaves. Front is 15.

Q023 PSEUDO CODE

[Data
Structures] After the operations, what does peek return?
Stack
S
push(S,
10)
push(S,
20)
push(S,
30)
pop(S)
push(S,
40)
Print
peek(S)

A 10
B 20
C 40
D 30

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Stack
after pops/pushes: [10, 20, 40]. peek = 40.

Q024 PSEUDO CODE

[Bitwise]
What is printed?
Integer
a, b
Set
a = 13, b = 6
Print
a & b

A 4
B 5
C 7
D 11

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
13
is 1101, 6 is 0110, AND is 0100 = 4.

Q025 PSEUDO CODE

[Recursion]
What does P(4) return?
Function
P(n)
if
n == 1 then return 2
return
P(n - 1) * 2
End
Function
Print
P(4)

A 8
B 12
C 16
D 24

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
P(1)=2,
P(2)=4, P(3)=8, P(4)=16.

Q026 PSEUDO CODE

What
does R(6) return?
Function
R(n)
if
n < 3 then return n
return
R(n - 1) + 2
End
Function
Print
R(6)

A 6
B 8
C 10
D 12

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
R(2)=2.
Each later call adds 2. R(6)=2 + 2×4 = 10.

Q027 PSEUDO CODE

What
is printed?
Stack
S
push(S,
3)
push(S,
6)
push(S,
9)
pop(S)
push(S,
12)
Print
peek(S)

A 3
B 6
C 12
D 9

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

After
pop, 9 is gone. 12 is pushed. peek returns 12.

Q028 PSEUDO CODE

What
is printed?
Integer
i, c
Set
c = 0
for
i = 1 to 8 do
if
i % 2 == 0 then c = c + i
end
for
Print
c

A 16
B 18
C 20
D 36

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Even
values 2+4+6+8 = 20.

Q029 PSEUDO CODE

What
is printed?
Integer
x
Set
x = 11
Print
x | 4

A 15
B 7
C 8
D 44

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
11
is 1011, 4 is 0100, OR is 1111 = 15.

Q030 PSEUDO CODE

What
is printed?
Queue
Q
enqueue(Q,
7)
enqueue(Q,
14)
enqueue(Q,
21)
dequeue(Q)
enqueue(Q,
28)
Print
front(Q)

A 7
B 14
C 21
D 28

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
7
is removed first. The new front is 14.

Q031 PSEUDO CODE

What
does U(4) return?
Function
U(n)
if
n == 0 then return 1
return
n + U(n - 1)
End
Function
Print
U(4)

A 8
B 9
C 11
D 24

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
U(0)=1,
then +1+2+3+4 = 11.

Q032 PSEUDO CODE

What
is printed?
Integer
z
Set
z = 14
Print
z & 7

A 6
B 7
C 8
D 21

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:

14
is 1110, 7 is 0111, AND is 0110 = 6.

Q033 PSEUDO CODE

What
is printed?
Integer
i, n
Set
n = 0
for
i = 1 to 6 do
if
i % 3 == 0 then n = n + i
end
for
Print
n

A 6
B 9
C 12
D 21

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

Multiples
of 3 from 1 to 6 are 3 and 6. Sum = 9.

Q034 PSEUDO CODE

What
is printed?
Queue
Q
enqueue(Q,
4)
enqueue(Q,
8)
enqueue(Q,
12)
dequeue(Q)
Print
front(Q)

A 4
B 8
C 12
D Empty

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
4
leaves first. Front is 8.

Q035 PSEUDO CODE

What
is printed?
Stack
S
push(S,
2)
push(S,
5)
push(S,
8)
pop(S)
push(S,
11)
Print
peek(S)

A 2
B 5
C 11
D 8

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Pop
removes 8. Push 11. peek returns 11.

Q036 PSEUDO CODE

What
is printed?
Stack
S
push(S,
6)
push(S,
9)
push(S,
15)
pop(S)
push(S,
21)
Print
peek(S)

A 6
B 9
C 21
D 15

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:
Pop
removes 15. Push 21. peek returns 21.

Q037 PSEUDO CODE

What
is printed?
Integer
k
Set
k = 21
Print
k >> 2

A 5
B 6
C 10
D 84

CORRECT ANSWER Option (A)

STEP-BY-STEP LOGIC & EXPLANATION:
21
is 10101. Right shift 2 gives 101 = 5.

Q038 PSEUDO CODE

What
is printed?
Integer
i, t
Set
t = 1
for
i = 1 to 5 do
t
= t * 2
end
for
Print
t

A 10
B 16
C 32
D 64

CORRECT ANSWER Option (C)

STEP-BY-STEP LOGIC & EXPLANATION:

Start
1, multiply by 2 five times: 32.

Q039 PSEUDO CODE

What
is printed?
Queue
Q
enqueue(Q,
3)
enqueue(Q,
9)
enqueue(Q,
15)
dequeue(Q)
Print
front(Q)

A 3
B 9
C 15
D Empty

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:

3
leaves. Front is 9.

Q040 PSEUDO CODE

What
does W(5) return?
Function
W(n)
if
n <= 1 then return 2
return
W(n - 1) + 3
End
Function
Print
W(5)

A 11
B 14
C 15
D 17

CORRECT ANSWER Option (B)

STEP-BY-STEP LOGIC & EXPLANATION:
W(1)=2.
Add 3 four times: 2+12=14.

Q041 PSEUDO CODE

What will be the
output of the following pseudo code
Integer
i, j, k, d
Set
j = 6, i = 8, k = 2, d = 1
if
((j ^ i) mod k equals 0) then
d = d + 1
else
d = d - 1
end
if
Print
d

A 1
B 3
C 0
D 2

CORRECT ANSWER Option (D)